5-MINUTE REVISION

Skewness and Kurtosis

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 7 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Skewness → degree of asymmetry of a distribution.
- Kurtosis → fourth standardised central moment, associated with tail behaviour and extreme observations.
- For a symmetric unimodal distribution with an existing mean and a unique central mode, mean = median = mode.
- Symmetric distribution → skewness = 0.
- Negatively skewed → longer left tail.

Quick revision · 2

- Many unimodal left-skewed distributions have mean < median < mode; this is not universal.
- Positively skewed → longer right tail.
- Many unimodal right-skewed distributions have mean > median > mode; this is not universal.
- Karl Pearson's measure → based on mean, mode or median, and standard deviation.
- Bowley's measure → based on quartiles.

Quick revision · 3

- Kelley's measure → based on percentiles, particularly P_{10}, P_{50}, and P_{90}.
- Moment-based skewness → based on the third central moment.
- β_1 → indicates the extent of skewness but not its direction.
- γ_1 → indicates both magnitude and direction of skewness.
- $\gamma_1 = 0$ → zero third-moment skewness; symmetry is not established.

Quick revision · 4

- $\gamma_1 > 0$ → positively skewed.
- $\gamma_1 < 0$ → negatively skewed.
- $\beta_2 = 3$ → mesokurtic.
- $\beta_2 > 3$ → leptokurtic.
- $\beta_2 < 3$ → platykurtic.

Quick revision · 5

- $\gamma_2 = 0$ → mesokurtic.
- $\gamma_2 > 0$ → leptokurtic.
- $\gamma_2 < 0$ → platykurtic.
- $\mu_1 = 0$ for every distribution.
- μ_2 is the population variance.

Quick revision · 6

- μ_3 is used for measuring skewness.
- μ_4 is used for measuring kurtosis.

Important formulae · 1

Karl Pearson's coefficient of skewness:

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}}$$

Alternative Karl Pearson's coefficient of skewness:

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\text{Standard deviation}}$$

Bowley's coefficient of skewness:

$$S_Q = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)}$$

Important formulae · 2

Alternative form of Bowley's coefficient:

$$S_Q = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

Kelley's coefficient of skewness:

$$S_P = \frac{(P_{90} - P_{50}) - (P_{50} - P_{10})}{(P_{90} - P_{50}) + (P_{50} - P_{10})}$$

rth central moment:

$$\mu_r = \frac{\sum_{i=1}^{k} f_i (x_i - \bar{x})^r}{N}$$

Important formulae · 3

Zeroth central moment:

$$\mu_0 = 1$$

First central moment:

$$\mu_1 = 0$$

Second central moment:

$$\mu_2 = \sigma^2$$

Important formulae · 4

Third central moment:

$$\mu_3 = \frac{\sum_{i=1}^{k} f_i (x_i - \bar{x})^3}{N}$$

Measure of skewness based on moments:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Karl Pearson's γ_1:

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \text{sgn}(\mu_3)\sqrt{\beta_1}$$

Important formulae · 5

Measure of kurtosis:

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

Kurtosis of excess:

$$\gamma_2 = \beta_2 - 3$$

Self-check from the chapter

1. The asymmetry of a distribution is measured by _____. **[Skewness]**
2. A distribution with a longer right tail is _____ skewed. **[Positively]**
3. A distribution with a longer left tail is _____ skewed. **[Negatively]**
4. Bowley's coefficient of skewness is based on _____. **[Quartiles]**

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